

AKSHAT YADAV

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EDUCATION

GL Bajaj Institute of Technology & Management

2025 – 2029

B.Tech, Computer Science & Engineering | Coursework: Data Structures & Algorithms, OOP, DBMS, OS, Networks

CGPA: 6.75 / 10.0

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript (ES6+), TypeScript, SQL, HTML5/CSS3

Backend & Systems: Node.js, Express.js, RESTful APIs, WebSockets, Next.js API Routes, UDP Sockets, JWT Auth, Microservices

Databases & Storage: PostgreSQL, Supabase (Row-Level Security), MongoDB, MySQL, DuckDB-WASM (In-Browser OLAP)

Tools & Cloud Platforms: Git, GitHub, Docker, Postman, Linux/Bash, Vercel, TailwindCSS, HTML5 Canvas API

Core Computer Science: Data Structures, Algorithms, Memory Management, System Architecture, Object-Oriented Programming (OOP)

FEATURED PROJECTS

ApexWall AI – Autonomous Sim Telemetry & Race Engineering Platform [\[Live Demo\]](#) [\[GitHub\]](#)

2026

Next.js 14, TypeScript, Node.js, Express.js, WebSockets, Supabase (PostgreSQL), DuckDB-WASM, Groq SDK

- **Engineered an in-browser telemetry engine** using `@duckdb/duckdb-wasm` in Web Workers, executing zero-latency SQL queries on multi-megabyte binary race files directly client-side without incurring backend server compute costs.
- **Built high-DPI HTML5 Canvas multi-channel scrubbers and 2D GPS circuit mappers** rendering 1:1 FIA coordinates, dynamic delta-time heatmaps, and Kamm's Friction Circle physics vectors at 60 FPS for precision driver coaching.
- **Implemented secure cloud persistence & versioning** with Supabase (PostgreSQL with Row-Level Security), enabling side-by-side parameter diffing between engineering setup revisions and public shareable URLs.
- **Integrated Groq LLaMA 3.3 70B inference** to synthesize corner-by-corner anomaly diagnosis (understeer/snap oversteer detection) into simulator-native chassis setup files (`.json`, `.sto`, `.svm`).
- **Formulated thermodynamic tyre pressure & fuel window solvers**, modeling atmospheric thermal gradients and dynamic fuel consumption stint windows with tactical lift-and-coast range extension math.

ApexWall Telemetry Bridge – High-Throughput UDP to WebSocket Gateway [\[GitHub\]](#)

2026

Node.js, Express.js, dgram (UDP Sockets), WebSockets (ws), Binary Buffer Parsing, Canvas HUD

- **Developed a standalone Node.js networking gateway** listening on UDP ports 9000 & 20777 to intercept high-frequency raw telemetry datagrams broadcasted by physics simulation engines (ACC, F1).
- **Implemented low-overhead binary buffer decoding**, converting raw byte streams into structured JSON telemetry payloads (RPM, speed, steering angle, G-forces) with sub-millisecond parsing latency.
- **Multiplexed packet streams to connected browser clients via WebSockets**, maintaining stable 60Hz frame delivery with automatic client reconnection, packet buffering, and heartbeat monitoring.
- **Built an interactive live cockpit dashboard** featuring a dynamic 15-LED shift light cluster, digital speedometer/tachometer, and real-time G-G crosshair vector tracking instantaneous grip limits.

ENGINEERING HIGHLIGHTS & PRACTICAL COMPETENCIES

- **Production Systems Deployment:** Engineered, containerized, and deployed full-stack web applications on Vercel with automated continuous deployment pipelines and live client traffic.
- **Real-Time Systems & Networking:** Hands-on experience developing low-latency network pipelines, asynchronous Node.js event loops, binary buffer parsing, and bidirectional WebSockets.
- **Database & Data Modeling:** Practical knowledge of relational database modeling (PostgreSQL), writing structured SQL schemas, Row-Level Security policies, and in-memory analytical queries.
- **Software Engineering Disciplines:** Proficient in Git/GitHub branching workflows, modular code structure, API lifecycle design, and responsive frontend integration.